TASK 4: HIGH-IMBALANCE CLINICAL DISEASE PREDICTOR PIPELINE

1. Introduction

Early identification of physiological abnormalities is an important application of machine learning in healthcare. Clinical datasets commonly contain an unequal number of healthy and disease-positive patients. This class imbalance can cause a machine-learning model to become biased toward the majority class and achieve apparently good accuracy while failing to identify a significant portion of patients who actually have the disease.

This project develops an end-to-end machine-learning classification pipeline for diabetes risk prediction using the Pima Indians Diabetes dataset. The objective was not simply to maximize overall accuracy, but to prioritize **recall, also known as sensitivity**, because the primary purpose of the model is to identify as many disease-positive patients as possible.

Three classification algorithms were developed and compared:

Support Vector Machine (SVM), Random Forest, and XGBoost.

Class imbalance was addressed using **Synthetic Minority Oversampling Technique (SMOTE)**. Hyperparameters were optimized using GridSearchCV with **5-fold stratified cross-validation**, while recall was used as the primary scoring metric.

The final pipeline also incorporated missing-value treatment, feature scaling where required, model evaluation, ROC-AUC analysis, confusion matrices, cross-validation recall arrays, and model persistence for future predictions.

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2. Dataset Description

The project used the **Pima Indians Diabetes dataset**, a binary classification dataset containing clinical measurements that can be used to estimate whether a patient demonstrates a diabetic pattern.

The loaded dataset contained **768 patient observations and 9 columns**, consisting of eight predictor variables and one target variable.

The input variables used in the project were:

Feature	Description
Pregnancies	Number of pregnancies
Glucose	Plasma glucose concentration
BloodPressure	Diastolic blood pressure

Feature	Description
SkinThickness	Triceps skin-fold thickness
Insulin	Serum insulin measurement
BMI	Body Mass Index
DiabetesPedigreeFunction	Diabetes hereditary/pedigree indicator
Age	Patient age
Outcome	Target class: 0 = negative, 1 = positive

The initial dataset technically contained no `NaN` values. However, inspection revealed a number of clinically unrealistic zero measurements. Specifically, there were 5 zero glucose values, 35 zero blood-pressure values, 227 zero skin-thickness values, 374 zero insulin values, and 11 zero BMI values.

These zero values were treated as missing measurements by replacing them with `NaN`. After this conversion, the corresponding missing-value counts became 5 for glucose, 35 for blood pressure, 227 for skin thickness, 374 for insulin, and 11 for BMI.

Median imputation was subsequently performed inside the machine-learning pipelines rather than before cross-validation. This design helps prevent information from validation folds from influencing the preprocessing of training folds.

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3. Class Imbalance Analysis

The target variable showed a clear imbalance between negative and positive cases.

The complete dataset contained:

Outcome	Number of Records	Percentage
No Diabetes – 0	500	65.10%
Diabetes – 1	268	34.90%

The distribution therefore consisted of approximately 65% negative and 35% positive observations.

Although this imbalance is not as extreme as some medical datasets, ignoring it could still bias the classifiers toward the majority class. This is particularly undesirable in a screening-oriented problem because false-negative predictions represent patients with the condition who are classified as negative.

For this reason, recall/sensitivity was selected as the main optimization criterion instead of accuracy.

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4. Data Preprocessing

The eight predictor variables were separated from the `Outcome` target. The resulting feature matrix contained 768 observations and eight input features.

An 80/20 stratified train-test split was then performed. The training dataset contained **614 samples**, while the independent testing dataset contained **154 samples**.

The training set consisted of 400 negative and 214 positive cases, while the test set consisted of 100 negative and 54 positive cases.

Stratification was important because it maintained approximately the same class ratio in both the training and testing datasets.

A `StratifiedKFold` cross-validation strategy with five folds, shuffling, and a random state of 42 was used throughout hyperparameter optimization and subsequent cross-validation evaluation.

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5. Handling Class Imbalance Using SMOTE

SMOTE was used to balance the training data.

Before SMOTE, the training set contained:

Class	Samples
Negative	400
Positive	214

After applying SMOTE, the training distribution became:

Class	Samples
Negative	400
Positive	400

The minority class was therefore synthetically increased from 214 observations to 400 observations.

An important design decision was to include SMOTE **inside each machine-learning pipeline** rather than applying it once before cross-validation.

The effective workflow was:

Imputation → optional scaling → SMOTE → classifier

This approach means that during cross-validation, SMOTE generates synthetic observations only from the training portion of each fold. It therefore reduces the risk of data leakage between the training and validation partitions.

For SVM, a `StandardScaler` was also included because SVM is sensitive to differences in feature magnitude. Random Forest and XGBoost did not require standard scaling.

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6. Classification Models and Hyperparameter Tuning

6.1 Support Vector Machine

The SVM pipeline consisted of median imputation, standard scaling, SMOTE, and an SVC classifier.

GridSearchCV tested combinations of the regularization parameter `C` and RBF-kernel gamma values. A total of 12 parameter combinations were evaluated across five cross-validation folds, resulting in 60 model fits.

The best SVM configuration was:

Hyperparameter	Selected Value
Kernel	RBF
C	0.5
Gamma	0.1

The tuned SVM achieved a best cross-validation recall of **0.7478**.

6.2 Random Forest

The Random Forest pipeline consisted of median imputation, SMOTE, and the Random Forest classifier.

Hyperparameters including the number of trees, maximum depth, minimum samples per split, minimum samples per leaf, and feature-selection strategy were tuned. Twenty-four parameter combinations were evaluated across five folds, producing 120 model fits.

The best Random Forest configuration was:

Hyperparameter	Selected Value
Number of estimators	200
Maximum depth	6
Maximum features	sqrt
Minimum samples per leaf	2
Minimum samples per split	2

The tuned Random Forest produced the highest mean cross-validation recall of **0.7524**.

6.3 XGBoost

The XGBoost pipeline contained median imputation, SMOTE, and an XGBoost classifier.

Parameters including learning rate, maximum tree depth, number of estimators, subsampling, and column sampling were optimized. Sixteen parameter combinations were evaluated across five folds, giving a total of 80 fits.

The best XGBoost configuration was:

Hyperparameter	Selected Value
Learning rate	0.03
Maximum depth	3
Number of estimators	150
Subsample	0.8
Column sample by tree	0.8

The optimized XGBoost model achieved a cross-validation recall of **0.7336**.

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7. Cross-Validation Recall Analysis

Recall was evaluated independently across the five stratified folds for each optimized classifier.

The observed recall arrays were:

Fold	SVM	Random Forest	XGBoost
1	0.7209	0.7209	0.7209
2	0.7907	0.7674	0.8140
3	0.6744	0.7442	0.6977
4	0.7674	0.7674	0.7209
5	0.7857	0.7619	0.7143
Mean	**0.7478**	**0.7524**	**0.7336**
Standard Deviation	**0.0442**	**0.0179**	**0.0411**

These values are taken directly from the executed cross-validation results.

Random Forest produced the highest mean recall at **75.24%**.

It also produced the lowest standard deviation, only **0.0179**, indicating that its recall performance was considerably more consistent between folds than SVM and XGBoost.

SVM performed similarly with a mean recall of 74.78%, but its standard deviation of 0.0442 demonstrated greater variation across folds.

XGBoost showed the lowest mean cross-validation recall at 73.36% and a standard deviation of 0.0411.

For a clinical prediction pipeline, this consistency is useful because a model that performs well only on particular data partitions may be less dependable when presented with previously unseen patients.

[Insert Figure: 5-Fold Cross-Validation Recall Box Plot here]

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8. Final Test-Set Results

After hyperparameter optimization, all three models were evaluated using the independent 154-sample test set.

Metric	SVM	Random Forest	XGBoost
Accuracy	**0.7727**	0.7468	0.7403
Precision	**0.6557**	0.6119	0.6029
Recall / Sensitivity	0.7407	**0.7593**	**0.7593**
Specificity	**0.7900**	0.7400	0.7300
F1 Score	**0.6957**	0.6777	0.6721
ROC-AUC	0.8148	0.8289	**0.8335**
PR-AUC	0.6954	**0.7277**	0.7224
Mean CV Recall	0.7478	**0.7524**	0.7336
CV Recall Std.	0.0442	**0.0179**	0.0411

The test results demonstrate that no single classifier dominated every evaluation measure.

SVM produced the highest overall accuracy at **77.27%**, the highest precision at **65.57%**, and the highest specificity at **79.0%**.

However, its sensitivity was 74.07%, meaning that it detected fewer positive diabetic cases than Random Forest and XGBoost.

Random Forest and XGBoost both produced a test recall of **75.93%**.

XGBoost produced the highest ROC-AUC at **0.8335**, suggesting strong overall discrimination between the two classes. Random Forest was close behind with a ROC-AUC of **0.8289**.

Random Forest also achieved the highest PR-AUC at **0.7277**, slightly exceeding XGBoost.

[Insert Figure: ROC Curve Comparison here]

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9. Confusion Matrix Analysis

The confusion matrices provide a more clinically interpretable representation of model behaviour.

Support Vector Machine

The SVM confusion matrix was:

	Predicted Negative	Predicted Positive
Actual Negative	79	21
Actual Positive	14	40

The model correctly identified 40 of the 54 positive patients while missing 14 positive cases.

Random Forest

The Random Forest confusion matrix was:

	Predicted Negative	Predicted Positive
Actual Negative	74	26
Actual Positive	13	41

Random Forest correctly identified 41 of the 54 positive cases and produced 13 false negatives.

XGBoost

The XGBoost confusion matrix was:

	Predicted Negative	Predicted Positive
Actual Negative	73	27
Actual Positive	13	41

Like Random Forest, XGBoost detected 41 of the 54 positive cases and missed 13.

This demonstrates the trade-off involved in improving clinical sensitivity. Random Forest and XGBoost detected one additional true-positive case compared with SVM, but they also generated more false-positive predictions.

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10. Final Model Selection

Random Forest was selected as the final model.

The decision was based primarily on cross-validation sensitivity rather than test accuracy alone.

The final Random Forest achieved:

Measure	Result
Test Accuracy	74.68%
Precision	61.19%
Recall / Sensitivity	**75.93%**
Specificity	74.00%
F1 Score	67.77%
ROC-AUC	82.89%
Mean 5-Fold CV Recall	**75.24%**
CV Recall Standard Deviation	**1.79%**

The notebook automatically selected Random Forest based on its mean cross-validation recall of 0.7524.

Although SVM achieved a higher test accuracy of 77.27%, accuracy was not the primary objective of the project. Its recall was lower, at 74.07%.

Similarly, XGBoost achieved the highest ROC-AUC of 83.35% and matched Random Forest's test recall, but its mean cross-validation recall was lower at 73.36% and less consistent across folds.

Random Forest therefore provided the strongest balance between positive-case detection and cross-validation stability for the stated objective.

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11. Clinical Interpretation

Recall is particularly important in an early-warning classification system.

Recall is calculated as:

Recall = TP / (TP + FN)

where TP represents correctly detected positive patients and FN represents patients with the condition who were incorrectly predicted as negative.

Random Forest achieved a recall of approximately **75.93%**, meaning that it correctly detected around three-quarters of the positive cases in the independent test set.

Its confusion matrix contained 41 true positives and 13 false negatives.

In comparison, SVM produced 40 true positives and 14 false negatives, while XGBoost produced the same positive-case detection result as Random Forest but demonstrated weaker cross-validation recall consistency.

However, the classifier should not be interpreted as a clinical diagnostic system. The output represents a statistical machine-learning classification based on the supplied measurements and training dataset.

Clinical decisions would require additional validation, larger and more representative datasets, appropriate medical governance, and evaluation by healthcare professionals.

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12. End-to-End Prediction Pipeline

The completed system followed the architecture:

Clinical data → invalid-value processing → stratified train/test split → median imputation → scaling where required → SMOTE → model training → hyperparameter optimization → stratified cross-validation → recall analysis → independent test evaluation → model selection → model persistence

The selected Random Forest model was exported successfully as:

`clinical_disease_predictor.pkl`

and the comparison results were saved as:

`task4_model_results.csv`.

A sample unseen patient was subsequently passed through the persisted prediction pipeline. The model returned class 1 with an estimated positive-class probability of **0.8373**, demonstrating that the complete pipeline could accept new raw patient measurements and generate an output.

This final step demonstrated that the work went beyond experimental model comparison and resulted in a reusable end-to-end machine-learning pipeline.

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13. Limitations and Future Improvements

Although the project successfully demonstrated imbalance-aware clinical classification, several limitations should be considered.

The dataset contains only 768 observations, which limits how confidently performance can be generalized to larger and more diverse populations. The class imbalance is meaningful but not extremely severe, so future work could evaluate the pipeline on datasets with substantially rarer positive outcomes.

SMOTE generates synthetic minority samples based on existing observations. Although this can improve model learning, synthetic observations do not replace genuinely collected patient data.

The current model also uses the default classification decision threshold. Future experimentation could optimize the probability threshold specifically for clinical sensitivity. Lowering the threshold could potentially detect additional positive cases, although this would usually increase the number of false positives.

Other extensions could include repeated stratified cross-validation, calibration analysis, feature-importance or SHAP explanations, Bayesian hyperparameter optimization, cost-sensitive classification, alternative oversampling techniques, and evaluation on external clinical datasets.

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14. Conclusion

This project successfully developed an end-to-end machine-learning pipeline for detecting diabetes-related physiological abnormalities in an imbalanced clinical dataset.

The workflow included data inspection, correction of physiologically invalid zero measurements, median imputation, stratified train-test splitting, SMOTE oversampling, model-specific preprocessing, hyperparameter optimization, five-fold stratified cross-validation, sensitivity-based evaluation, confusion-matrix analysis, ROC-AUC evaluation, and model persistence.

Three machine-learning algorithms were evaluated: SVM, Random Forest, and XGBoost.

SVM achieved the highest test accuracy at **77.27%**, while XGBoost produced the highest ROC-AUC at **83.35%**. However, the project objective emphasized reliable identification of disease-positive patients rather than maximizing accuracy.

Random Forest achieved **75.93% test sensitivity** and the highest mean cross-validation recall of **75.24%**, with the lowest recall variation across folds at only **0.0179 standard deviation**.

For this reason, Random Forest was selected as the final model.

Overall, the experiment demonstrates why accuracy alone is insufficient for imbalanced medical classification. Incorporating SMOTE, stratified cross-validation, hyperparameter optimization, and recall-focused evaluation produced a more appropriate pipeline for an early abnormality detection scenario.

The final model should be considered a machine-learning research and educational prototype rather than a validated medical diagnostic system.

Internship Task: Task 4 – Mini Project
Project: High-Imbalance Clinical Disease Predictor Pipeline
Primary Evaluation Metric: Recall / Sensitivity
Selected Model: Random Forest
Final Test Recall: 75.93%
Mean 5-Fold CV Recall: 75.24%
ROC-AUC: 82.89%